

Shohei Yoshida

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SUMMARY

Incoming Master of Engineering student at the School of Design and Science, Chiba Institute of Technology (Apr 2026). Senior at the University of Tsukuba with an interdisciplinary background in International Studies and Computer Science. Possess a strong analytical foundation from both humanities research methodologies and foundational science/engineering coursework. Focus on leveraging this dual perspective to pursue innovative research in Probabilistic Computing.

EDUCATION

Chiba Institute of Technology, Narashino, Chiba, Japan

- Master of Engineering, School of Design and Science (Expected Mar 2028) Apr 2026 – Present

University of Tsukuba, Tsukuba, Ibaraki, Japan

- Bachelor of Arts in International Development Apr 2022 – Mar 2026
 - College of International Studies
 - Cumulative GPA: 3.32

RESEARCH PROJECTS

- **Bachelor's Thesis: A Static-Graph Acceleration Framework for CrossCat in GenJAX** University of Tsukuba, 2026
 - Reformulated CrossCat's nonparametric Bayesian model into a static computational graph compatible with JAX/GenJAX, addressing scalability limitations arising from dynamically changing CRP-based cluster structures.
 - Introduced truncated Stick-Breaking Process (SBP) approximations to replace CRP representations, enabling fixed-size tensor representations for column-view and row-cluster assignments.
 - Reimplemented collapsed Gibbs sampling using JAX vmap and JIT compilation, eliminating Python-level loops and enabling efficient parallel execution.
 - Achieved up to $11.3\times$ speedup on the 10,000-row Adult dataset and up to $4.7\times$ speedup for long MCMC runs (5,000 iterations), while preserving dependency recovery accuracy (ROC-AUC ≈ 1.0).
 - Supervised by Prof. Keisuke Kameyama.
- **Bayesian Mixture-of-Hills for Marketing Mix Modelling** Ongoing
 - Developed a Bayesian mixture-of-Hill-curves framework to model heterogeneous consumer response in marketing mix analysis.
 - Extended traditional single-curve marketing mix models to a Dirichlet-weighted mixture formulation, allowing latent response segments and saturation behaviours to be inferred from data.
 - Implemented the model in NumPyro and JAX, enabling GPU-accelerated posterior inference.
 - Evaluated model performance on synthetic and real-world datasets using WAIC and ELPD-LOO, demonstrating improved uncertainty calibration under heterogeneous response patterns with limited overhead in homogeneous settings.
 - Analyzed identifiability and label-switching issues inherent in mixture models and assessed posterior diagnostics (R-hat, ESS).
- **Adaptive Data Acquisition via Information Theory for GenJAX Programs** 2025
 - Applying and exploring generic entropy and information theory-based estimation techniques within probabilistic programs (GenJAX) to enable more efficient and adaptive data acquisition strategies.
 - Focus: Probabilistic Computing (GenJAX), Information Theory, Entropy Estimation, Bayesian Optimization.

PRESENTATIONS

- **Privacy-Preserving Synthetic Data Generation using Probabilistic Computing** Jul 2025
 - Poster presentation at the Symposium of Design and Science 2025, Chiba Institute of Technology, Tokyo, Japan.
 - Shohei Yoshida, Grisha Szep, Mizuki Oka.
 - Proposed a privacy protection method using CrossCat, a nonparametric Bayesian model, to selectively remove statistical dependencies between sensitive variables by constraining its view structure. Demonstrated controllable and transparent privacy-utility trade-offs on synthetic datasets, with inference accelerated up to $\sim 700\times$ via JAX/GenJAX.

CERTIFICATIONS

- Fundamental Information Technology Engineer Examination Mar 2025

LANGUAGES

- Japanese: Native language.
- English: Fluent (speaking, reading, writing; TOEIC L&R: 865).

COMPUTER SKILLS

- German: Basic (speaking, reading, writing).
- **Programming Languages:** Python, C++, SQL, Julia
- **Frameworks & Libraries:** JAX, NumPyro, pandas, scikit-learn, Gen
- **Tools & Platforms:** GCP, Git, GitHub, Linux, SQLite, Docker, $\LaTeX$

WORK EXPERIENCE

Japan Business Art inc., Tokyo, Japan

- Student Intern Mar 2023 – Oct 2023
 - Analyzed past internal corporate newsletters to identify client companies' internal communication challenges and areas for improvement.
 - Contributed to the planning, proposal, and design of internal newsletter content and layouts intended to address client companies' internal issues.
 - Edited and refined internal newsletter content to improve clarity and engagement.
 - Supported the creation of Investor Relations (IR) materials through research, content proposal, and editorial review.

CAMPUS ACTIVITIES

University of Tsukuba Tea Ceremony Club, University of Tsukuba

- Treasurer, Web and SNS Manager Jan 2023 – Dec 2023
 - As Treasurer, responsible for creating and managing the club's budget and finances.
 - Organized and managed tea ceremony events.
 - Maintained the club's official webpage and managed its social media presence.
 - Participated in the 33rd Urasenke Student Seminar hosted by the Urasenke Tankokai General Headquarters.

[CV compiled on 2026-05-07]